AI-enhanced Flipped CLIL Models for ESP Education: A Systematic Review in Tourism and Hospitality Contexts

Abstract

This study presents a systematic review of empirical research examining the integration of Artificial Intelligence (AI), flipped classroom strategies, and Content and Language Integrated Learning (CLIL) in English for Specific Purposes (ESP) instruction, with a particular focus on tourism and hospitality contexts. Guided by the PRISMA 2020 guidelines and the PSALSAR framework, peer-reviewed journal articles published between 2016 and 2024 were systematically identified, screened, and synthesized from major academic databases. A total of 18 empirical studies met the inclusion criteria and were included in the final analysis. The findings indicate that ESP instruction grounded in needs-based pedagogy is most effectively supported when AI, flipped learning, and CLIL operate as complementary pedagogical components. AI primarily functions as a pedagogical enabler by supporting personalization, formative feedback, and learner autonomy, while flipped classroom strategies restructure learning activities toward interactive and task-based engagement. CLIL further facilitates the integration of language development with domain-specific knowledge essential for professional communication. Notably, the synthesis suggests that these approaches collectively form an instructional ecology, in which instructional relevance, learning activities, and technological support are interdependently aligned. Despite promising outcomes, the review reveals a limited number of studies that holistically integrate all four dimensions. This study, therefore, provides a consolidated pedagogical perspective and identifies directions for future research on AI-enhanced flipped CLIL models in ESP education. This review contributes by conceptualizing AI-enhanced flipped CLIL in ESP education as an instructional ecology, in which Artificial Intelligence, flipped classroom pedagogy, and CLIL function as interdependent pedagogical mechanisms rather than isolated techniques.